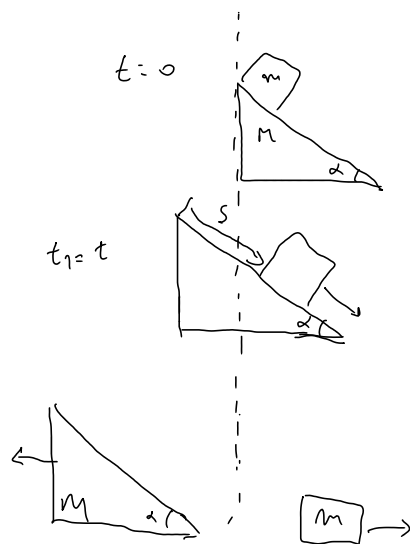




$$E = mgl \sin \alpha$$

$$A_x = 0, A_y = 0, a_x = 0, a_y = 0, a_m = 0$$

$$E = \frac{1}{2} M V_x^2 + \frac{1}{2} m (v_x^2 + v_y^2) + (mgl \sin \alpha - mgh)$$

$$A_x, a_x, a_y, a_m$$

$$E = \frac{1}{2} M V_x^2 + \frac{1}{2} m (v_x^2 + v_y^2)$$

$$A_x, a_x, a_y = 0, a_m = 0, A_y = 0$$

i) Conservação de momento

$$\vec{p}_M^{(s)} + \vec{p}_m^{(s)} = \vec{p}_M^{(a)} + \vec{p}_m^{(a)}$$

$$\vec{p}_M^{(0)} = \vec{p}_m^{(0)} = 0 \Rightarrow M V_x + m v_x = 0$$

$$\Rightarrow M A_x + m a_x = 0$$

$$\begin{cases} -R \sin \alpha = M A_x \\ R \sin \alpha = m a_x \end{cases}$$

$$\begin{cases} F - R \sin \alpha = M A_x \\ R \sin \alpha = m a_x \end{cases}$$

$$\Rightarrow F = M A_x + m a_x$$

$$\Rightarrow \int_0^t F dt = M \int_0^t A_x dt + m \int_0^t a_x dt$$

$$\Rightarrow F \cdot \Delta t = M V_x + m v_x$$

$\Rightarrow$

